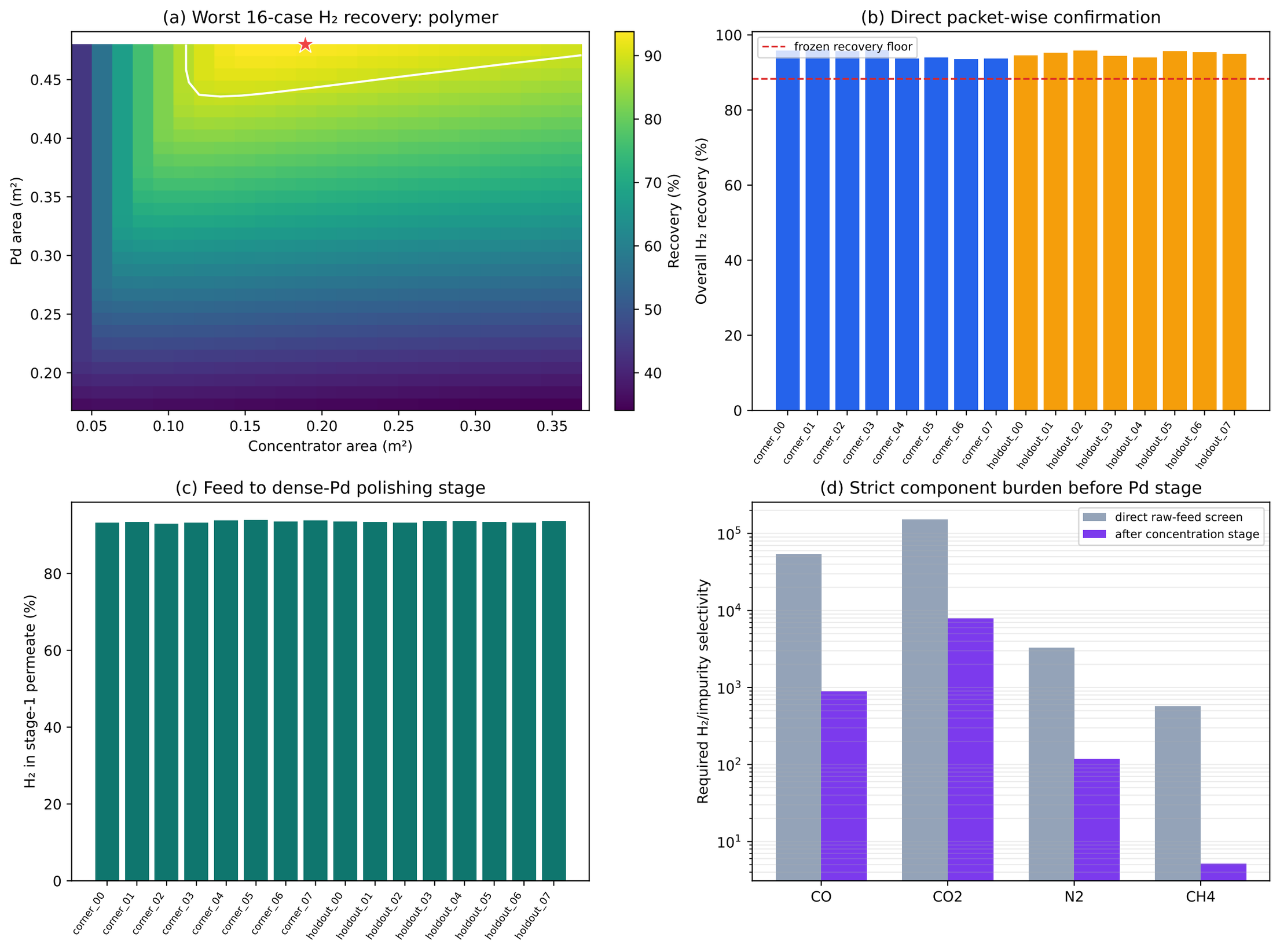
(a) Worst 16-case H₂ recovery: polymer
Pd area (m²)
Concentrator area (m²)
Recovery (%)
(b) Direct packet-wise confirmation
Overall H₂ recovery (%)
frozen recovery floor
corner_00
corner_01
corner_02
corner_03
corner_04
corner_05
corner_06
corner_07
holdout_00
holdout_01
holdout_02
holdout_03
holdout_04
holdout_05
holdout_06
holdout_07
(c) Feed to dense-Pd polishing stage
H₂ in stage-1 permeate (%)
(d) Strict component burden before Pd stage
Required H₂/impurity selectivity
direct raw-feed screen
after concentration stage
CO
CO2
N2
CH4